REPRESENTABLE MORPHISMS ARE CLOSED UNDER COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1014-1081, following the statement at 1007-1012.

Editorial note. The uniqueness argument spells out both uses of the universal property explicitly.

Allowed standard prerequisites (editorial).

- Fibre products and their universal property. Definition definition-fibre-products of this chapter, used at proof lines 1030 and 1072. The existence halves give the two factorization diagrams; the uniqueness halves give both the uniqueness of delta and its use in closing the final step.

Source reference. Original definition 001V: sources/stacks-categories/pinned-original-source.tex, source lines 933–933. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let $f : x \to y$, and $g : y \to z$ be representable. Then $g \circ f : x \to z$ is representable.*

Proof. Let $t \in \mathrm{Ob}(\mathcal{C})$ and $\varphi \in \mathrm{Mor}_{\mathcal{C}}(t, z)$. As g and f are representable, we obtain commutative diagrams

$$\begin{array}{ccc} y \times_z t & \xrightarrow{q} & t \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle \varphi} \\ y & \xrightarrow{g} & z \end{array} \qquad \begin{array}{ccc} x \times_y (y \times_z t) & \xrightarrow{q'} & y \times_z t \\ {\scriptstyle p'}\downarrow & & \downarrow{\scriptstyle p} \\ x & \xrightarrow{f} & y \end{array}$$

with the universal property of Definition 001V. We claim that $x \times_z t = x \times_y (y \times_z t)$ with morphisms $q \circ q' : x \times_z t \to t$ and $p' : x \times_z t \to x$ is a fibre product. First, it follows from the commutativity of the diagrams above that $\varphi \circ q \circ q' = g \circ f \circ p'$. To verify the universal property, let $w \in \mathrm{Ob}(\mathcal{C})$ and suppose $\alpha : w \to x$ and $\beta : w \to t$ are morphisms with $\varphi \circ \beta = g \circ f \circ \alpha$. By definition of the fibre product, there are

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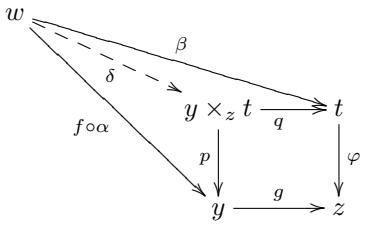
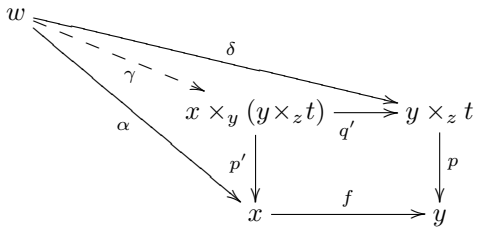
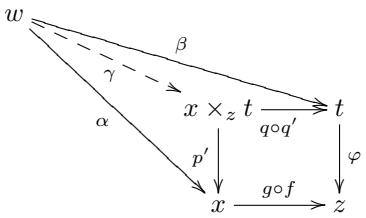
unique morphisms δ and γ such that



and



commute. Then, γ makes the diagram



commute. To show its uniqueness, let γ' verify $q \circ q' \circ \gamma' = \beta$ and $p' \circ \gamma' = \alpha$. Because γ is unique, we just need to prove that $q' \circ \gamma' = \delta$ and $p' \circ \gamma' = \alpha$ to conclude. We supposed the second equality. For the first one, we also need to use the uniqueness of delta. Notice that δ is the only morphism verifying $q \circ \delta = \beta$ and $p \circ \delta = f \circ \alpha$. We already supposed that $q \circ (q' \circ \gamma') = \beta$. Furthermore, by definition of the fibre product, we know that $f \circ p' = p \circ q'$. Therefore:

$$p \circ (q' \circ \gamma') = (p \circ q') \circ \gamma' = (f \circ p') \circ \gamma' = f \circ (p' \circ \gamma') = f \circ \alpha.$$

Then $q' \circ \gamma' = \delta$, which concludes the proof. □